MICAEL SANTANA

SENIOR SOFTWARE ENGINEER | JAVA SPECIALIST (QUARKUS & SPRING) & ANGULAR

📍 Ribeira do Pombal - BA, Brazil 📞 +55 (75) 9 9946-8431 ✉️ micaelsantana.dev@gmail.com

🌐 micaelsantana.netlify.app 💻 github.com/iSantanaaH

PROFESSIONAL SUMMARY

Senior Software Engineer with **6+ years of experience** architecting high-performance distributed systems, multitenant SaaS platforms, and reactive backend architectures. Extensive specialization in the Java ecosystem (Quarkus and Spring Boot) alongside event-driven and reactive platforms utilizing RabbitMQ, Redis, PostgreSQL, and Keycloak. On the frontend, I possess solid experience building scalable micro-frontends and Single Page Applications using Angular, RxJS, and modular architectures. Strong expertise in Domain-Driven Design (DDD), Clean Architecture, fault-tolerance resilience patterns, and automated CI/CD deployment pipelines on Kubernetes.

CORE COMPETENCIES & TECHNICAL STACK

Backend & Reactive Systems

Java 21, Quarkus (Reactive/Mutiny), Spring Boot 3, RESTful APIs, GraphQL, Keycloak IAM, OAuth2/OIDC, OpenSSL, JWT Rotation, Hibernate / Panache.

Distributed Systems & Messaging

Event-Driven Architecture (EDA), RabbitMQ (DLX, AMQP), Redis (Rate-Limiting & Cache), Resilience Patterns, Circuit Breakers, Multitenant Data Isolation.

Frontend & Micro-frontends

Angular 14–22+, TypeScript, RxJS, Angular CLI, Workspaces & Nx, Reactive Forms, State Management, TailwindCSS, SCSS, Web Performance Optimization.

DevOps, Cloud & Observability

Docker, Kubernetes (K8s), Helm, Horizontal Pod Autoscaler (HPA), GitHub Actions CI/CD, Prometheus, Grafana, SonarQube, Clean Code, TDD.

PROFESSIONAL EXPERIENCE

Senior Full Stack Software Engineer & Architect | MW-Flow (Commercial SaaS Enterprise)

Sep/2026 – Present

- Technical leadership in designing and architecting the MW-Flow multitenant SaaS ecosystem, built under a reactive microservices architecture using Quarkus (Java 21) and asynchronous messaging.
- Modeled multitenant database isolation strategies in PostgreSQL combined with execution context resolution via `TenantResolver`, ensuring complete data separation across organizational tenants.
- Implemented the centralized identity and federated authentication microservice (`ms-auth`) using Keycloak (OAuth2/OpenID Connect) and defensive request rate-limiting backed by Redis.
- Engineered an Offline-First Point of Sale (POS) solution (`ms-pos`) featuring ultra-resilient local sync queues, store-level contingency, and async invoicing integrations with SEFAZ (NFC-e/NF-e).
- Architected the decoupled E-Commerce platform (`ms-ecommerce`), handling digital payment gateway integrations (Pix/Credit Card) and automated inventory reservation/deduction via RabbitMQ events.
- Constructed a modular Angular workspace (`mw-flow-web`) housing three target applications (`web-app`, `pos-app`, `ecommerce-app`) focused on web performance, lazy loading, and component reusability.

`Java 21` `Quarkus` `Angular 22+` `Keycloak` `RabbitMQ` `Redis` `PostgreSQL` `Kubernetes`

Angular Frontend Developer | Kinyx Tecnologia

2021 – 2022

- Developed and maintained enterprise-grade web applications of high complexity using Angular, TypeScript, HTML5, SCSS, and reactive streams with RxJS.
- Created internal libraries of reusable UI components and shared services, boosting standardization and overall development delivery velocity across engineering teams.
- Integrated Single Page Applications (SPA) with backend RESTful APIs, optimizing data fetching, global error handling, and visual user experience feedback.
- Engineered web performance optimizations (reducing initial bundle loading times, leveraging OnPush Change Detection strategies, and ensuring universal responsiveness).
- Collaborated in agile environments (Scrum/Kanban) adhering to strict Code Review guidelines and structured Git versioning workflows.

`Angular` `TypeScript` `RxJS` `REST APIs` `SCSS` `Git`

ARCHITECTURAL SHOWCASE & REFERENCE PROJECT

MW-Flow | Multitenant SaaS Enterprise System (POS & ERP)

Active / Public Portfolio

High-performance distributed platform designed for omnichannel retail operations. Demonstrates end-to-end orchestration of 4 decoupled Quarkus microservices, offline POS contingency support, event-driven messaging via RabbitMQ, and modular Angular frontends.

- **Decoupled Blueprint Strategy:** Maintained a public macro-architecture repository (`mw-flow-architecture`) detailing C4 specifications, OpenAPI 3.0 contracts, and multitenant system diagrams, keeping proprietary core logic and infrastructure in private repositories.
- **Observability & Governance:** Built automated CI/CD pipelines (GitHub Actions) integrated with SonarQube for static code analysis, code coverage, and vulnerability management, alongside Prometheus/Grafana stacks for real-time Kubernetes cluster telemetry.

`Quarkus` `Clean Architecture` `Event-Driven` `GraphQL` `GitHub Actions` `Prometheus & Grafana`

EDUCATION

Bachelor's / Associate Degree in Software Analysis and Development

Completed (2022 – 2024)

UNOPAR - Universidade Norte do Paraná

LANGUAGES & ENGINEERING STANDARDS

Languages:

- Portuguese: Native / Fluent
- English: Full Professional Proficiency (Advanced technical writing & reading)
- Spanish: Basic

Engineering Governance:

- Domain-Driven Design (DDD) & Clean Architecture
- SOLID Principles, Design Patterns & Code Refactoring
- OWASP Top 10 Security Guidelines & Identity IAM